

# Statement of Purpose

## Prologue

As the valedictorian of the Electrical Engineering program at Hanoi University of Science and Technology (HUST), Vietnam's most prestigious engineering institution, I am applying to the PhD program in Electrical Engineering at University of Wisconsin–Madison (UW–Madison). Through four years of rigorous training and extensive research engagement, I have built a strong foundation spanning power electronics, control, optimization, and hardware-software co-design. My long-term goal is to become an independent researcher and professor advancing the stability and control of modern power systems, and UW–Madison's exceptional research environment offers the ideal platform to pursue this vision.

## Professional Plans

My interest in power and energy systems began with a simple curiosity: *how do complex electrical systems behave when scaled, interconnected, or pushed to their limits?* As I engaged more deeply with power electronics, control, and system dynamics, I realized that contributing meaningfully to this field requires far more than mastering coursework, it demands rigorous research, creativity, and the ability to translate theory into practical solutions.

My immediate goal is to pursue a PhD focused on the modeling, control, and stability of future power systems with high penetration of inverter-based resources. In the long term, I envision a career in academia, leading a research group that explores the emerging challenges of power-electronic-dominated grids. I aspire to mentor students, build collaborative research teams, and contribute to technologies that support reliable, sustainable, and resilient power infrastructure worldwide.

To reach this goal, I seek a research environment where faculty mentorship, interdisciplinary collaboration, and hands-on experimentation are integral. UW–Madison's strong ecosystem in power electronics, electric power systems, and control aligns perfectly with this vision.

## Research Interests

During my undergraduate years, I developed two interconnected research passions that now define my academic identity.

First, I became captivated by energy management and system optimization in microgrids. While my formal coursework in linear algebra, calculus, and numerical methods provided the mathematical foundation, I developed my optimization skills primarily through hands-on research and lab work. Working on real-world energy management problems, I learned to formulate and solve complex coordination challenges, an experience that taught me far more than textbooks alone could offer.

Second, I developed a deep interest in the dynamic behavior and stability of inverter-based power systems. As renewable sources increasingly penetrate the grid, understanding their dynamic interactions becomes critical. I found immense satisfaction in using mathematical models to predict system behavior rather than relying solely on experiments. To pursue this passion, I sought opportunities beyond typical undergraduate requirements, developing expertise in modeling and converter dynamics through both industry internship and academic research.

These two threads, system-level management and device-level dynamics, now shape my core research identity. I aim to bridge analytical modeling with practical implementation to develop robust, scalable, and high-performance solutions for modern power systems.

## My Research Experiences and Preparations

My research journey reflects a deliberate progression toward becoming an independent researcher capable of bridging theory and practice across the full stack, from mathematical modeling to hardware implementation.

Early in my undergraduate years, my internship at INS Engineering under the guidance of Dr. Tuan Ngo, Principal Engineer, opened my eyes to industrial-scale power system modeling. Working with PSCAD and MATLAB/Simulink, I modeled grid-connected inverter dynamics and motor control algorithms. The challenge was not just running simulations; it was making them accurate down to the smallest waveform details. Through systematic parameter identification, I learned to translate physical systems into mathematical representations that matched reality, a skill that proved foundational for everything that followed.

The real test came when I led HUST's first hydrogen-based microgrid prototype, a project demanding ownership across the entire engineering stack. I designed the system architecture coordinating PV, electrolyzer, fuel cell, battery, and loads, then developed control algorithms and embedded them into Arduino-based firmware under the supervision of Prof. Nguyen Duc Tuyen. At the circuit level, I hand-crafted necessary power converter modules: selecting circuit components, laying out PCBs,

soldering, debugging hardware-software interactions. This taught me that robust systems emerge only when you understand every layer, control theory guides algorithms; however, microcontroller constraints and hardware non-idealities demand adaptation. Leading a team under tight deadlines also taught me project orchestration, delegation, and maintaining momentum through setbacks. Our prototype was showcased at exhibitions across Vietnam and recognized at an international conference in Japan.

My time in Taiwan marked different growth. At National Chung Cheng University's Power Quality Research Lab, I shifted from broad integration to specialized depth, focusing on harmonic mitigation in grid-connected inverters under the guidance of Prof. Gary W. Chang. I designed a PR-HC controller with impedance shaping, then tuned it through small-signal modeling and experimental iteration. Reducing inverter THD from 12.41% to 3.14%, below IEEE standards 5.00% for low-voltage distribution grid, reinforced a conviction: elegant mathematical models, properly validated, yield dramatic real-world improvements.

These experiences transformed me from executing assigned tasks to driving projects from conception through validation. More importantly, they taught me how research actually works: collaborating with senior researchers, communicating setbacks honestly, supporting junior teammates, and persisting when results do not come easily. These are precisely the qualities I bring to your university.

### **Why I Am a Strong Fit**

UW–Madison's ECE department offers the ideal environment for me to pursue my research vision, and I am particularly drawn to the work of Professor Dominic Gross. His research on power systems stability and dynamic modeling of inverters directly addresses what I am most passionate about. I am drawn to understanding power systems from their fundamental mathematical foundations - how differential equations and impedance interactions govern stability, and how rigorous analytical frameworks can predict system behavior. Leveraging my experience in inverter control and experimental validation, I aim to contribute meaningfully to the development and analysis of inverter-dominated power systems.

Beyond Professor Gross's group, I am also keen to work with Professors Giri Venkataramanan and Manish K. Singh, whose complementary expertise closely aligns with my academic background. Professor Venkataramanan's research on power converters would strengthen my foundation in converter-level modeling, control, and hardware-oriented design. In parallel, Professor Singh's work in smart grid, optimization and control would deepen my understanding of system-level coordination, stability, and decision-making in power systems. Together, these viewpoints would allow me to develop a cohesive research approach that bridges device-level converter behavior and grid-level dynamics, equipping me to address emerging challenges in advanced energy systems with both theoretical rigor and practical relevance.

### **Closing**

I bring a strong combination of hardware skills, software proficiency, analytical depth, and research maturity shaped by early exposure to real-world engineering challenges. I work well in collaborative environments, value mentorship, and hold myself to high standards in meeting deadlines, executing projects, and continually seeking better methods and solutions. I am committed to dedicating my energy, curiosity, and discipline to the pursuit of impactful research.

Thank you for considering my application. I look forward to contributing to and learning from UW–Madison's distinguished research community.